Page 1: Document Control & Engineering Metadata

Document Identifier: HLD-PT-MASTER-2026-V1.0
System Domain: Electric Powertrain Domain Controller (ECU_EMS_01)
Classification: Tier-1 Automotive Engineering Release

Target Platform: AUTOSAR Classic Platform Release 4.4.0
Microcontroller: Infineon AURIX TC397XX (3x TriCore 300MHz with Lockstep)
Operating System: AUTOSAR OS Scalability Class 4 (SC4)

Revision History:
- Rev 0.1 (Jan 2026): Initial architectural decomposition.
- Rev 0.5 (Apr 2026): Added CAN FD communication matrices.
- Rev 0.9 (Jul 2026): Integrated ISO 26262 ASIL B/C safety allocations.
- Rev 1.0 (Sep 2026): Production Release baseline.

Authorized Sign-Offs:
- Chief Architecture Lead: Dr. E. Schmidt
- Functional Safety Manager: H. Weber (ISO 26262 Certified)
- Cybersecurity Officer: V. Rossi (ISO/SAE 21434 Lead)

Page 2: Table of Contents & Specification Index

Table of Contents Summary:

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Page 3: Executive System Summary & Scope

Executive Overview:
This High-Level Design specification defines the production software architecture for the Electric Powertrain Domain Controller (ECU_EMS_01).

The controller is responsible for pedal interpretation, multi-source torque arbitration, traction motor inverter control, thermal management, high-voltage interlock monitoring, and on-board diagnostics.

Page 4: Applicable Industry Standards & Norms

Normative References:
- AUTOSAR Classic Platform Release 4.4.0 (R20-11)
- ISO 26262:2018 Road Vehicles - Functional Safety (Parts 1 to 12)
- ISO/SAE 21434:2021 Road Vehicles - Cybersecurity Engineering
- ISO 14229-1:2020 Road Vehicles - Unified Diagnostic Services (UDS)
- ISO 11898-1:2015 Road Vehicles - Controller Area Network (CAN with Flexible Data-Rate)
- MISRA C:2012 Guidelines for the use of the C language in critical systems

Page 5: Engineering Stakeholders & Organizational RACI

Stakeholder Matrix:
- Vehicle Architecture Team: Responsible for vehicle topology and high-level requirements.
- Powertrain Software Team: Accountable for SWC implementation and RTE mapping.
- Safety Assessment Board: Consulted for ASIL B and ASIL C decomposition sign-offs.
- Validation & Calibration Team: Informed for HIL and dynamometer test execution.

Page 6: Target Microcontroller Specification (Infineon AURIX TC397)

Microcontroller Core Architecture:
- Processor: Infineon AURIX TC397XX 32-bit Microcontroller.
- Cores: 3x TriCore CPUs running at 300 MHz (Core 0, Core 1, and Core 2).
- Lockstep Protection: Core 0 and Core 1 feature dedicated hardware lockstep comparator cores for SIL-3 / ASIL D hardware fault coverage.
- Memory: 16 MB on-chip Flash with ECC, 2.9 MB SRAM with parity check.

Hardware Safety Features:
- On-chip Memory Protection Unit (MPU) with 32 configurable regions.
- Hardware Security Module (HSM) supporting AES-128, AES-256, and ECC-256 cryptographic hardware acceleration.

Page 7: Power Supply & Voltage Regulation Architecture

System Power Management IC (PMIC):
- Controller: Infineon TLF35584 Safety PMIC with integrated window watchdog.
- Battery Input: 9V to 16V nominal (4.5V cold-crank survival, 36V load dump protection).
- Regulated Rails: +5.0V Analog, +3.3V Digital I/O, +1.25V Core Logic.

Power-Down Hold Time: Hardware capacitance guarantees minimum 50ms hold-up after KL15 ignition-off for clean NvM EEPROM shutdown.

Page 8: Analog Sensor Acquisition Hardware Channels

Analog Input Channel Assignment:
- ADC0_CH1: Accelerator Pedal Position Sensor 1 (0.5V to 4.5V).
- ADC0_CH2: Accelerator Pedal Position Sensor 2 (0.25V to 2.25V - Cross-plausibility).
- ADC1_CH4: High-Voltage DC Link Voltage Measurement (0 to 1000V via attenuator).
- ADC1_CH5: Traction Inverter Phase U Temperature Thermistor (NTC).
- ADC2_CH3: 12V Auxiliary Battery Sense Voltage.

Page 9: Digital Inputs and Frequency Capture Channels

Digital I/O & Timer Subsystem:
- GTM (Generic Timer Module): Handles microsecond-resolution crankshaft and wheel speed pulse capture.
- DigIn_01: Brake Pedal Switch Primary (Hall-effect normally open contact).
- DigIn_02: Brake Pedal Switch Secondary (Mechanical switch normally closed contact).
- DigIn_03: High-Voltage Interlock Loop (HVIL) physical continuity monitor.

Page 10: Actuator Output Drivers & H-Bridge Stages

Power Output Drivers:
- Out_HBridge_01: PWM driver for electronic throttle valve positioning (20 kHz switching).
- Out_LowSide_01: Coolant circulation pump relay control (2A clamp).
- Out_HighSide_01: Main contactor pre-charge control output.

Page 11: Inverter SPI Interface & Gate Driver Topology

Inverter Gate Driver Interface:
- Primary Interface: High-Speed Synchronous SPI (10 Mbps) connecting to Infineon EiceDRIVER gate driver ICs.
- Read Channel: Real-time IGBT desaturation fault detection within 1.5 microseconds.
- Write Channel: Programmable phase dead-time and active short-circuit (ASC) safety commands.

Page 12: Vehicle Communication Physical Transceivers

CAN FD Transceivers:
- Transceiver 1 (Drivetrain CAN): NXP TJA1044GT high-speed CAN FD transceiver.
- Transceiver 2 (Diagnostic CAN): NXP TJA1044GT with autonomous bus wake-up.
- Common Mode Chokes: 51 uH choke on CANH/CANL lines for strict OEM EMC Class 5 compliance.

Page 13: Electromagnetic Compatibility (EMC) Architecture

EMC Design Measures:
- Ground Plane: 4-layer PCB with dedicated internal ground plane separating digital logic from power switching stages.
- ESD Protection: Transient Voltage Suppressors (TVS) rated for 25 kV air discharge on all connector pins.
- Filtering: LC low-pass filters on analog power rails.

Page 14: Connector Pinout & Wire Harness Assignment

ECU Master Connector Pinout (56-Pin Automotive Grade):
- Pin 1-4: Power GND.
- Pin 5: KL30 Battery Permanent (+12V).
- Pin 6: KL15 Ignition Switched (+12V).
- Pin 12-13: Powertrain CAN FD (CANH_A / CANL_A).
- Pin 14-15: Diagnostic CAN FD (CANH_B / CANL_B).
- Pin 22-25: Throttle Actuator H-Bridge Motor Leads.

Page 15: Hardware Watchdog & Safety Management

Hardware Watchdog Protocol:
- External PMIC TLF35584 monitors micro-controller health via Challenge-Response Watchdog (SPI token exchange every 10ms).
- If the software fails to respond with the correct mathematical response within 2 consecutive cycles, the PMIC asserts a physical reset line, cutting inverter gate drive within 500 microseconds.

Page 16: Layered Architecture Principles & Methodology

AUTOSAR Layering Strategy:
1. Application Layer: Reusable Software Components (SWCs) encapsulating control algorithms.
2. Runtime Environment (RTE): Abstract communication middleware generated from ARXML descriptors.
3. Basic Software (BSW): Standardized system services, memory, and bus protocols.
4. Microcontroller Abstraction Layer (MCAL): Hardware peripheral access routines.

Page 17: Runtime Environment (RTE) Contract Phase

RTE Generation Concepts:
- Contract Phase: Defines component header files (Rte_<SWC>.h) with port API prototypes.
- Generation Phase: Generates concrete buffer mappings, OS event triggers, and inter-core memory transfers.
- Zero Direct Pointer Passing: All data exchanged between SWCs is strictly routed through RTE APIs (Rte_Read, Rte_Write, Rte_Call).

Page 18: Sender-Receiver Interface Architecture

Sender-Receiver (S/R) Communication:
- Paradigm: 1-to-N asynchronous data broadcast.
- Implicit S/R: RTE copies data buffer at runnable start to ensure data consistency during execution.
- Explicit S/R: Direct API invocation returning Std_ReturnType (RTE_E_OK, RTE_E_LOST_DATA).
- Buffer Protection: Inter-task buffer updates protected via Os Interrupt Locking (SuspendOSInterrupts).

Page 19: Client-Server Interface Architecture

Client-Server (C/S) Communication:
- Paradigm: Remote procedure call (RPC) pattern.
- Synchronous Call: Calling runnable blocks until server runnable returns result (used for Fast Math routines).
- Asynchronous Call: Non-blocking invocation where client polls for result via Rte_Result API (used for DEM diagnostic reporting).

Page 20: Basic Software (BSW) Module Decomposition

BSW Core Stack Configuration:
- System Services: EcuM, BswM, OS, WdgM.
- Communication Services: Com, PduR, CanIf, CanSm, CanNm.
- Memory Services: NvM, Fee, Fls.
- Diagnostic Services: Dcm, Dem, FiM.
- MCAL Drivers: Gpt, Adc, Port, Dio, Spi, Can, Mcu.

Page 21: Inter-Core Communication (IOC) Architecture

Multi-Core Data Sharing:
- Mechanism: AUTOSAR OS Inter-OS-Application Communication (IOC).
- Lockless Queues: Single-reader single-writer circular queues allocated in shared on-chip LMU SRAM.
- Cross-Core Latency: Maximum transfer latency between Core 1 and Core 0 measured at 2.4 microseconds.

Page 22: Memory Protection Unit (MPU) Region Allocation

Memory Partitioning & Isolation:
- OS Application Core 0: Non-Safety BSW Stack (Privileged Mode).
- OS Application Core 1: ASIL B / ASIL C Torque Control Algorithms (User Mode with strict MPU bounds).
- OS Application Core 2: QM Diagnostic Reporting and Memory (User Mode).
- MPU Violation Handler: Triggers OS ShutdownHook and safely transitions inverter to free-wheeling state.

Page 23: ECU State Management (EcuM) Sequences

ECU Lifecycle Sequence:
1. EcuM_Init(): Hardware clock, PLL, and internal RAM initialization.
2. EcuM_StartupTwo(): Starts AUTOSAR OS and initializes BSW modules.
3. EcuM_GoToRun(): All periodic OS tasks activated.
4. EcuM_GoToSleep(): Prepares memory, saves non-volatile parameters, enters low-power stop mode.

Page 24: BSW Mode Manager (BswM) Rule Engine

BswM State Arbitration Rules:
- Rule 1: When KL15 == TRUE and BatteryVoltage > 9.0V, request Communication Mode COMM_FULL_COMMUNICATION.
- Rule 2: When OverTemperatureFault == TRUE, request DeratedTorqueMode.
- Rule 3: When CrashDetected == TRUE, execute EmergencyShutdownAction.

Page 25: Watchdog Manager (WdgM) Supervision Entities

Software Program Flow Monitoring:
- Alive Supervision: Monitors that 10ms Torque Task executes between 9.5ms and 10.5ms.
- Deadline Supervision: Checks that Throttle Actuator calculation finishes within 1.2ms of trigger.
- Logical Supervision: Verifies that torque calculation steps execute in valid sequential order (A -> B -> C).

Page 26: ISO 26262 Item Definition & Safety Goals

Powertrain Item Definition:
The Powertrain Controller is an electrical/electronic system responsible for translating driver inputs into vehicle motive force.

Safety Goals (SG):
- SG-01: Prevent unintended vehicle acceleration (ASIL B / ASIL C in v2.0).
- SG-02: Prevent unintended deceleration exceeding 0.3g (ASIL B).
- SG-03: Prevent loss of electrical isolation and high-voltage exposure (ASIL C).
- SG-04: Prevent unintended motor reverse torque while in forward drive gear (ASIL B).

Page 27: Hazard Analysis and Risk Assessment (HARA)

HARA Classification Matrix:
- Scenario H-01: Vehicle traveling at highway speeds (Exposure E4), driver pedal sensor shorts to ground (Severity S3), driver controllability (Controllability C2) -> Result: ASIL B.
- Scenario H-02: Sudden unintended regenerative brake lockup on icy road (S3, E3, C3) -> Result: ASIL C.
- Scenario H-03: Inverter short-circuit causing sudden loss of propulsion (S2, E4, C1) -> Result: ASIL A.

Page 28: Functional Safety Requirements (FSR) Allocation

Derived Safety Requirements:
- FSR-01: TorqueArbitrationSWC shall arbitrate driver demand using cross-plausible dual sensor inputs within 50ms.
- FSR-02: In the event of dual-sensor disagreement exceeding 8%, torque demand shall be clamped to 0 Nm within 30ms.
- FSR-03: Vehicle speed broadcasts must be validated with End-to-End CRC checks before processing.

Page 29: Technical Safety Requirements (TSR) & Architecture

Technical Safety Concepts:
- Plausibility Check: Dual-channel independent ADC conversion for accelerator pedal potentiometer 1 and 2.
- Core Lockstep: Torque calculations executed on lockstep TriCore Core 1.
- Safe State: Open all inverter phase switches, de-energize throttle motor, and illuminate driver cluster telltale.

Page 30: Freedom From Interference (FFI) Proof

FFI Implementation:
1. Spatial Isolation: HW Memory Protection Unit (MPU) prevents QM tasks from writing to ASIL B buffers.
2. Temporal Isolation: Fixed-priority preemptive OS scheduling with execution budget monitoring (WdgM).
3. Communication Isolation: Sender-Receiver buffer duplication managed exclusively by the generated RTE.

Page 31: Fault Tolerant Time Interval (FTTI) Calculations

FTTI Budgets:
- Unintended Acceleration FTTI: 100 milliseconds.
  * Sensor Acquisition Time: 10ms.
  * Algorithm Execution & Cross-Check: 10ms.
  * Actuator Transmission & Reaction: 35ms.
  * Total System Fault Reaction Time: 55ms (< 100ms budget, PASS).

Page 32: Safety Mechanism: Dual Accelerator Sensor Plausibility

Pedal Sensor Plausibility Algorithm:
Sensor 1 output range: 0.5V (0%) to 4.5V (100%).
Sensor 2 output range: 0.25V (0%) to 2.25V (100%).
Mathematical Constraint: Output(Sensor 1) - 2 * Output(Sensor 2) must remain < 0.15V.
If violated for > 4 consecutive 10ms cycles, DTC 0x921401 is declared.

Page 33: Safety Mechanism: Torque Plausibility Model

Level 2 Torque Cross-Check:
A secondary mathematical model running on Core 0 compares arbitrated motor torque against maximum vehicle acceleration limits derived from vehicle wheel speeds. If demand exceeds physical traction capacity by 25%, safe state is triggered.

Page 34: Safety Critical Calibration Parameter Protection

Calibration Safety Measures:
Safety-related calibration parameters (e.g., maximum motor torque limit: 1200 Nm) are stored in duplicate inverted Flash memory locations. The RTE verifies integrity via CRC32 on every ECU startup before enabling torque output.

Page 35: Safety Verification and Validation Plan

Validation Testing Requirements:
- Fault Injection Testing: Injecting short-circuits, stuck-at-ground, and CAN bus bit-flips on HIL simulator.
- Structural Code Coverage: 100% Statement and Branch Coverage, MC/DC (Modified Condition/Decision Coverage) for ASIL B/C modules.

Page 36: Cybersecurity Management System & ISO 21434 Scope

Automotive Cybersecurity Objectives:
Compliance with ISO/SAE 21434:2021 and UNECE World Forum R155 regulations.
Ensures vehicle drive-by-wire torque commands cannot be spoofed, intercepted, or replayed by unauthorized diagnostic or telematics tools.

Page 37: Threat Analysis & Risk Assessment (TARA)

TARA Assessment Findings:
- Threat T-01: Spoofing of CAN torque request frame (0x1A4) via compromised telematics box.
  * Attack Vector: Remote Cellular / CAN Bus Gateway.
  * Impact: Safety Critical (Vehicle unwanted torque).
  * Feasibility: Medium.
  * Risk Score: High -> Requires SecOC cryptographic authentication.

Page 38: Secure Onboard Communication (SecOC) Specification

SecOC Protocol Configuration:
- Target Message: CAN_Powertrain_TorqueDemand (0x1A4).
- Algorithm: AES-128 CMAC (Cipher-based Message Authentication Code).
- Truncated MAC Length: 24 bits appended to CAN FD payload.
- Freshness Value: 64-bit Trip Counter + 8-bit Cyclic Reset Counter.

Page 39: Hardware Security Module (HSM) Integration

AURIX HSM Micro-controller Subsystem:
- Dedicated ARM Cortex-M3 security processor isolated from main TriCore CPUs.
- Stores Root of Trust, Secure Boot verification keys, and SecOC symmetric pre-shared keys.
- Key Access: Main CPU cannot read private keys; keys are invoked only via cryptographic hardware handles.

Page 40: Secure Boot & Firmware Authentication Chain

Secure Boot Sequence:
1. Primary Bootloader (ROM) verifies Secondary Bootloader (Flash) using RSA-3072 signature.
2. Secondary Bootloader computes SHA-256 hash of AUTOSAR Application image.
3. If signature verification succeeds within 25ms, execution transfers to EcuM_Init.

Page 41: Diagnostic Security Access (UDS Service 0x27)

Cryptographic Security Access:
- Security Level 1 (Calibration Access): 64-bit random seed generated via hardware True Random Number Generator (TRNG).
- Security Level 9 (Flashing Access): Asymmetric ECDSA signature verification with NIST P-256 curve.

Page 42: Intrusion Detection System (IDS) Sensor Integration

CAN Bus IDS Monitoring Rules:
- Sensor IDS_CAN_Rate: Flags anomaly if CAN_Powertrain_TorqueDemand is received faster than 7ms (Bus flooding attack).
- Sensor IDS_DLC_Check: Flags payload length mismatches.
- Anomaly Action: Log security incident to internal persistent audit memory.

Page 43: Secure Flash Programming & Firmware Over-the-Air (FOTA)

FOTA Integrity Requirements:
All firmware updates delivered over-the-air must be encrypted with AES-256-GCM and signed by the OEM Root CA. Downgrade attacks are blocked via an anti-rollback hardware monotonic counter.

Page 44: Software Asset Protection & Trust Boundaries

Trust Boundaries Defined:
- Trust Boundary 1: External Diagnostic Port (Untrusted).
- Trust Boundary 2: Gateway ECU SOME/IP Routing (Partially Trusted).
- Trust Boundary 3: Internal Lockstep Core 1 Execution (Fully Trusted Core Zone).

Page 45: Vulnerability Management & Incident Response

Cybersecurity Lifecycle Procedures:
Annual penetration testing, continuous SBOM (Software Bill of Materials) CVE scanning, and rapid firmware security patch deployment protocol compliant with ISO/SAE 21434 Section 13.

Page 46: Application Software Component Taxonomy

Component Inventory Overview:
This chapter specifies the core software components executing on the Powertrain Controller:
- TorqueArbitrationSWC (Section 6.1)
- EngineSpeedSensorSWC (Section 6.2)
- ThrottleActuatorSWC (Section 6.3)
- CruiseControlSWC (Section 6.4)
- ThermalManagementSWC (Section 6.5)
- HighVoltageMonitorSWC (Section 6.6)
- RegenerativeBrakingSWC (Section 6.7)

Page 47: TorqueArbitrationSWC Detailed Architecture

Component: TorqueArbitrationSWC (ApplicationSWC)
Purpose: Central vehicle torque demand calculation, driver pedal filter, and slip limitation.
Execution: 10ms periodic task (OsTask_10ms_Core1), ASIL B (Upgraded to ASIL C in v2.0).

Configured Ports:
- Rp_DriverDemand (R-Port / Required): Interface If_DriverDemand
- Rp_EngineSpeed (R-Port / Required): Interface If_EngineSpeed
- Pp_TorqueCommand (P-Port / Provided): Interface If_TorqueCommand

Page 48: TorqueArbitrationSWC Internal Runnables

Runnable Entities:
1. RE_TorqueArbitration_Init(): Executed on startup to initialize internal integrator states to zero.
2. RE_TorqueArbitration_10ms(): Periodic execution runnable. Implements driver pedal map lookup, rate-of-change limiter, and cruise control torque arbitration.

Page 49: TorqueArbitrationSWC Data Dictionary

Data Elements:
- AccelPedalPosition_pct: uint8 (0 to 100%, Resolution: 0.5%/bit).
- TargetTorque_Nm: sint16 (-500 to +1200 Nm, Resolution: 0.1 Nm/bit).
- TorqueValid: boolean (TRUE = Normal Operation, FALSE = Fallback Active).
- DriverKickdownSwitch: boolean (TRUE = Full throttle detent switch closed).

Page 50: EngineSpeedSensorSWC Detailed Architecture

Component: EngineSpeedSensorSWC (SensorActuatorSWC)
Purpose: Digital signal acquisition of crankshaft rotation from magnetic pickup sensor.
Execution Rate: 10ms, ASIL B.

Ports:
- Pp_EngineSpeed (Provided): Interface If_EngineSpeed (EngineSpeed_rpm: 0..8000 RPM).
- Pp_SensorDiag (Provided): Interface If_SensorDiag.

Page 51: EngineSpeedSensorSWC Processing Algorithm

Signal Processing Logic:
Raw pulse intervals captured via GTM timer are filtered using an exponential moving average: RPM_filtered = 0.7 * RPM_current + 0.3 * RPM_previous. Detects missing tooth sync pulse every 360 degrees.

Page 52: ThrottleActuatorSWC Detailed Architecture

Component: ThrottleActuatorSWC (SensorActuatorSWC)
Purpose: Position closed-loop PID control of electronic throttle butterfly valve.
Execution Rate: 10ms, ASIL B.

Ports:
- Rp_TorqueCommand (Required): Interface If_TorqueCommand.
- Pp_ActuatorState (Provided): Interface If_ActuatorFeedback (ThrottleAngle_deg: 0..900).
- Pp_ActuatorFault (Provided): Interface If_ActuatorFaultReport.

Page 53: ThrottleActuatorSWC Fail-Safe Logic

Actuator Fail-Safe Behavior:
If TorqueValid == FALSE or CAN communication timeout is detected, the H-bridge is completely de-energized. A dual mechanical return spring snaps the throttle plate to a default 7.5-degree limp-home opening.

Page 54: CruiseControlSWC Detailed Architecture

Component: CruiseControlSWC (ApplicationSWC)
Purpose: Longitude vehicle cruise speed regulation and adaptive distance limiting.
Execution Rate: 20ms, ASIL B.

Ports:
- Rp_DriverCruiseSwitches (Required): Interface If_CruiseSwitches.
- Pp_CruiseTorqueLimit (Provided): Interface If_CruiseTorque.

Page 55: ThermalManagementSWC Detailed Architecture

Component: ThermalManagementSWC (ApplicationSWC)
Purpose: Inverter power stage temperature monitoring and coolant pump PWM command.
Execution Rate: 100ms, QM.

Ports:
- Rp_InverterTemps (Required): Interface If_ThermalSensors.
- Pp_PumpCommand (Provided): Interface If_CoolantPump (PumpDutyCycle_pct: 0..100%).

Page 56: ThermalManagementSWC Derating Strategy

Thermal Derating Curves:
- Below 105°C: 100% torque available.
- 105°C to 125°C: Linear torque reduction to 60%.
- Above 135°C: Emergency power reduction to 20% to prevent Silicon Carbide (SiC) MOSFET thermal destruction.

Page 57: HighVoltageMonitorSWC Detailed Architecture

Component: HighVoltageMonitorSWC (SensorActuatorSWC)
Purpose: Reads traction battery voltage, current, and High-Voltage Interlock (HVIL) physical line.
Execution Rate: 20ms, ASIL B.

Ports:
- Rp_BMS_Limits (Required): Interface If_BatteryPowerLimits.
- Pp_HVIL_Status (Provided): Interface If_HVIL_State (HVIL_Closed: boolean).

Page 58: RegenerativeBrakingSWC (v2.0 Enhancement)

Component: RegenerativeBrakingSWC (ApplicationSWC - Introduced in v2.0)
Purpose: Computes electric negative torque during braking for energy recovery.
Execution Rate: 5ms, ASIL C.

Ports:
- Rp_BatteryState (Required): Interface If_BatteryStatus.
- Pp_RegenTorqueLimit (Provided): Interface If_RegenTorque.

Page 59: TractionControlSWC Detailed Architecture

Component: TractionControlSWC (ApplicationSWC)
Purpose: Monitors wheel slip difference across driven wheels and limits positive torque to maintain lateral stability on slippery road surfaces.
Execution Rate: 10ms, ASIL B.

Page 60: InverterGateDriverCDD Complex Device Driver

Complex Device Driver: CDD_InverterGateDriver
Bypasses standard RTE to provide direct register access to AURIX High-Speed SPI for 50-microsecond inverter fault notification and phase desaturation shutdown.

Page 61: Inter-Component Data Flow & Connection Matrix

Signal Traceability Matrix:
- DriverDemandSWC:Pp_DriverDemand -> TorqueArbitrationSWC:Rp_DriverDemand
- EngineSpeedSensorSWC:Pp_EngineSpeed -> TorqueArbitrationSWC:Rp_EngineSpeed
- TorqueArbitrationSWC:Pp_TorqueCommand -> ThrottleActuatorSWC:Rp_TorqueCommand

Page 62: Software Component Memory Budget Allocations

RAM & ROM Usage Limits:
- TorqueArbitrationSWC: ROM 48 KB, RAM 4.2 KB.
- ThrottleActuatorSWC: ROM 32 KB, RAM 2.1 KB.
- EngineSpeedSensorSWC: ROM 18 KB, RAM 1.0 KB.
- Total Application Layer Footprint: ROM 412 KB, RAM 28 KB.

Page 63: RTE Calibration Parameters (Calibration Parameters)

Calibration Handles:
- Cal_MaxTorqueLimit_Nm: 1200 Nm default (Protected).
- Cal_PedalDeadband_Pct: 2.5% default.
- Cal_CruiseMaxSpeed_Kmh: 180 km/h default.

Page 64: Software Component Unit Testing Specifications

Component Test Requirements:
Each SWC shall be tested with 100% branch and statement coverage in a simulated RTE test harness before integration into the BSW build pipeline.

Page 65: Component Sign-Off & Verification Status

Verification Declaration:
All software component definitions documented in this chapter have undergone architectural static analysis and peer review.

Page 66: In-Vehicle Networking Architecture & Topologies

Vehicle Bus Topology:
Powertrain ECU communicates over two CAN FD channels:
- Powertrain CAN FD (CAN0): 500 kbps nominal, 2.0 Mbps data phase.
- Body CAN Gateway (CAN1): 500 kbps standard CAN.
- Ethernet Connection (100BASE-T1): Future domain controller link.

Page 67: CAN Controller Driver (Can) & Hardware Configuration

CAN Driver Settings:
- Controller: AURIX MultiCAN+ CAN0.
- Nominal Bit Timing: 500 kbps (Bit length: 2000ns, Sample Point: 80%).
- Fast Bit Timing: 2000 kbps (Bit length: 500ns, Sample Point: 75%).
- Transceiver Delay Compensation (TDC): Enabled.

Page 68: CAN Interface (CanIf) Layer Routing

CanIf Configuration:
Maps raw CAN hardware message buffers (HTH/HRH) to upper BSW modules: PduR, CanNm, and CanSm.
Configured with 32 Rx hardware mailboxes and 16 Tx priority-driven mailboxes.

Page 69: PDU Router (PduR) Routing Matrix

PduR Routing Paths:
- PDU CAN_Powertrain_TorqueDemand: Routed from Com module to CanIf.
- PDU UDS_Diag_Phys_Req: Routed from CanIf to Dcm module.
- PDU UDS_Diag_Phys_Resp: Routed from Dcm to CanIf.

Page 70: COM Stack Configuration & Signal Unpacking

AUTOSAR COM Module:
Handles packing and unpacking of signal data types (uint8, sint16) into network byte order (Motorola Big-Endian / Intel Little-Endian) with sign extension.

Page 71: CAN Message CAN_Powertrain_TorqueDemand (0x1A4)

Frame Specification: CAN_Powertrain_TorqueDemand
- CAN Frame Identifier: 0x1A4 (11-bit Standard ID).
- Transmission Period: 10 milliseconds (Cyclic).
- Payload Length: 16 Bytes (CAN FD).
- Signals:
  * Byte 0-1: TargetTorque_Nm (sint16, -500..+1200 Nm, Intel Endian).
  * Byte 2: TorqueValid (boolean, bit 0).
  * Byte 7: E2E_Counter (4-bit, 0..14).
  * Byte 8: E2E_CRC8 (8-bit CRC polynomial 0x1D).

Page 72: CAN Message CAN_EngineSpeed_Broadcast (0x2B0)

Frame Specification: CAN_EngineSpeed_Broadcast
- CAN Frame Identifier: 0x2B0.
- Transmission Period: 20 milliseconds.
- Payload Length: 8 Bytes.
- Signals: EngineSpeed_rpm (Byte 0-1), SpeedValid (Byte 2).

Page 73: CAN Message CAN_BatteryStatus_Rx (0x3C0)

Frame Specification: CAN_BatteryStatus_Rx
- Frame Identifier: 0x3C0 (Received from external BMS).
- Reception Timeout: 50 milliseconds.
- Signals: BatterySoC_pct (Byte 0), MaxChargePower_kW (Byte 1-2).

Page 74: End-to-End (E2E) Communication Library Overview

E2E Protection Concept:
Mandatory safety library protecting against bus transmission errors (lost messages, repeated messages, corrupted data, delayed frames).

Page 75: E2E Profile P01 Specification (10ms Messages)

E2E Profile P01 Configuration:
- Message ID: 0x1A4 (CAN_Powertrain_TorqueDemand).
- Data ID: 0x12.
- CRC Polynomial: 0x1D (SAE-J1850).
- Counter: 4-bit (0..14).
- Reception Max Jitter: 35ms (3 missed periods).

Page 76: E2E Profile P02 Specification (5ms Messages in v2.0)

E2E Profile P02 Configuration (Upgraded in v2.0):
- Message ID: 0x1A4.
- CRC: 16-bit CRC (CCITT 0x1021).
- Counter: 8-bit alive counter (0..255).
- Designed for high-frequency 5ms electric drive control.

Page 77: CAN Network Management (CanNm) Protocol

CanNm State Machine:
- Repeat Message State: 150ms broadcast on network activity.
- Normal Operation State: Periodic 100ms NM PDU broadcast.
- Ready Sleep State: Stops NM transmission; transitions to bus sleep if silent for 2000ms.

Page 78: CAN State Manager (CanSm) & Bus-Off Recovery

Bus-Off Recovery Strategy:
When Transmit Error Counter (TEC) > 255, CAN controller enters Bus-Off.
- Quick Recovery: 5 attempts spaced 20ms apart.
- Slow Recovery: After 5 failures, attempts spaced 1000ms apart to prevent bus disruption.

Page 79: Transport Layer (CanTp) ISO 15765-2 Protocol

CanTp Multi-Frame Diagnostics:
Supports segmentation of diagnostic payloads up to 4095 bytes (Single Frame, First Frame, Consecutive Frame, Flow Control).

Page 80: Communication Stack Timing & Latency Budgets

Timing Guarantees:
- Maximum Software Ingestion Latency: 1.2ms.
- Worst-Case Bus Queuing Delay: 2.8ms.
- Total Transmit Delay: < 5.0ms for 10ms periodic messages.

Page 81: Diagnostic Communication Manager (Dcm) Overview

UDS Subsystem Architecture:
The DCM handles ISO 14229-1 request dispatching, session arbitration, security verification, and response formatting.

Page 82: Diagnostic Session Control (Service 0x10)

Supported Diagnostic Sessions:
- 0x01: DefaultSession (Normal vehicle operation).
- 0x02: ProgrammingSession (Firmware bootloader flashing).
- 0x03: ExtendedDiagnosticSession (Sensor calibration and actuator testing).

Page 83: ECU Reset (Service 0x11)

Reset Options:
- Sub-function 0x01: HardReset (Power-cycle simulation).
- Sub-function 0x03: SoftReset (Restart application software without cutting peripheral power).

Page 84: Clear Diagnostic Information (Service 0x14)

DTC Erasure Protocol:
- Command 0x14 FFFFFF: Clears all confirmed and pending DTC fault records in NVRAM memory.

Page 85: Read DTC Information (Service 0x19)

DTC Query Services:
- Sub-function 0x02: ReportDTCByStatusMask (Find active DTCs).
- Sub-function 0x04: ReportSnapshotRecordByDTCNumber (Read freeze frame data).

Page 86: Read Data By Identifier (Service 0x22) Specification

Service 0x22 Implementation:
Allows diagnostic scan tools to read live engineering variables, sensor voltages, and system status registers.

Page 87: Data Identifier (DID) Master Catalogue (Part 1)

Primary Vehicle DIDs:
- DID 0xF190: Vehicle Identification Number (VIN) - 17 Bytes ASCII.
- DID 0xF188: Vehicle Manufacturer ECU Software Number - 'SW-EMS-TC397-V1.0'.
- DID 0xF195: System Supplier ECU Software Version Number - 'V01.00.04'.

Page 88: Data Identifier (DID) Master Catalogue (Part 2)

Powertrain Sensor DIDs:
- DID 0x2100: Current Arbitrated Torque Output (sint16, 0.1 Nm/bit).
- DID 0x2105: Accelerator Pedal Position Live Value (uint8, 0..100%).
- DID 0x2110: Engine Rotational Speed (uint16, 0.25 RPM/bit).
- DID 0x2120: Throttle Valve Angle Position (uint16, 0.1 deg/bit).
- DID 0x2201: Regenerative Torque Arbitrated Level (sint16, introduced in v2.0).

Page 89: Security Access (Service 0x27) Cryptographic Handshake

Security Access Sequence:
1. Client requests seed: 0x27 0x01.
2. ECU generates 64-bit random seed using on-chip TRNG.
3. Client returns 64-bit cryptographic key: 0x27 0x02 <Key>.
4. If valid, security level unlocked; if 3 failed attempts, 10-minute lockout imposed.

Page 90: Routine Control (Service 0x31) & Calibration Writing

Supported Routines:
- RID 0x0200: Calibrate Electronic Throttle Zero-Endstop Position.
- RID 0x0210: High-Voltage Inverter Self-Test Sequence.

Page 91: Diagnostic Event Manager (Dem) Subsystem Overview

DEM Subsystem Purpose:
Manages diagnostic events reported by BSW and SWCs, controls DTC status bytes, qualifies faults, and stores freeze frames in non-volatile memory.

Page 92: DTC Status Byte State Transitions (ISO 14229)

DTC Status Bit Mask:
- Bit 0: testFailed
- Bit 1: testFailedThisOperationCycle
- Bit 2: pendingDTC
- Bit 3: confirmedDTC
- Bit 4: testNotCompletedSinceLastClear
- Bit 5: testFailedSinceLastClear
- Bit 6: testNotCompletedThisOperationCycle
- Bit 7: warningIndicatorRequested

Page 93: DTC 0x921401: Throttle Position Circuit Fault

Fault Code: DTC 0x921401
- Fault Description: Throttle Valve Position Mismatch / Circuit Voltage High.
- Debounce Type: Counter-based (Step-up: 2, Step-down: 1, Threshold: 20).
- Immediate Action: Clamp torque request to maximum 50 Nm limp-home mode.
- Cluster Indicator: Check Engine Malfunction Indicator Lamp (MIL) illuminated.

Page 94: DTC 0x921502: Engine Speed Signal Plausibility Fault

Fault Code: DTC 0x921502
- Fault Description: Crankshaft Hall Sensor Signal Rate of Change Exceeds Physical Limit.
- Debounce Type: Time-based (150ms continuous failure).
- Immediate Action: Switch to secondary wheel speed calculation estimate.

Page 95: DTC 0x922004: CAN_Powertrain_TorqueDemand E2E Fault

Fault Code: DTC 0x922004
- Fault Description: E2E Profile P01 CRC Error or Alive Counter Failure on Bus.
- Debounce Type: 3 consecutive corrupted frames.
- Immediate Action: Invalidate received torque command; default to 0 Nm safe stop.

Page 96: DTC 0x932001: Regenerative Torque Mismatch (v2.0)

Fault Code: DTC 0x932001 (New in Revision 2.0)
- Fault Description: Regenerative Braking Coordination Mismatch / Battery Overcharge Limit Violation.
- Immediate Action: Disengage regenerative braking; rely 100% on hydraulic foundation brakes.

Page 97: Freeze Frame & Snapshot Data Recording

Freeze Frame Data Record 0x01 Content:
When any confirmed DTC is set, the following environmental parameters are frozen into NVRAM:
- Vehicle Speed (DID 0x2102)
- Battery Voltage (DID 0x2104)
- Odometer Mileage (DID 0x2108)
- Inverter Temperature (DID 0x2112)

Page 98: Function Inhibition Manager (FiM) Inhibit Rules

FiM Inhibit Logic:
- If DTC 0x921401 == CONFIRMED: Inhibit Function FID_CruiseControl and FID_LaunchControl.
- If DTC 0x932001 == CONFIRMED: Inhibit Function FID_RegenBraking.

Page 99: Fault Debouncing Algorithms (Counter & Time-based)

Debouncing Architecture:
- Counter-based: Used for fast electrical checks (prevents single-sample noise glitches).
- Time-based: Used for thermal and mechanical system diagnostics requiring settled physical states.

Page 100: DTC Aging and Fault Memory Healing

Healing Criteria:
If an operating cycle completes without the fault recurring, the confirmed DTC is aged.
After 40 fault-free warm-up driving cycles, the DTC is automatically erased from NVRAM fault memory.

Page 101: Operating System (OS) Scheduling Architecture

AUTOSAR OS Configuration:
Scalability Class 4 (SC4) providing memory protection, execution time budgeting, and priority-driven preemptive task scheduling across 3 TriCore CPU cores.

Page 102: OS Task Allocation Matrix (Core 1 - Safety)

Core 1 Task Mapping (ASIL B / ASIL C):
- OsTask_5ms_Core1 (Priority 22): RegenerativeBrakingSWC (v2.0).
- OsTask_10ms_Core1 (Priority 15): TorqueArbitrationSWC, ThrottleActuatorSWC, EngineSpeedSensorSWC.
- OsTask_20ms_Core1 (Priority 10): CruiseControlSWC.

Page 103: OS Task Allocation Matrix (Core 0 - Communication)

Core 0 Task Mapping (BSW & Networking):
- OsTask_1ms_Core0 (Priority 30): CanIf_MainFunction, Spi_MainFunction.
- OsTask_10ms_Core0 (Priority 20): Com_MainFunction_Tx, Com_MainFunction_Rx, PduR_MainFunction.
- OsTask_20ms_Core0 (Priority 12): HighVoltageMonitorSWC.

Page 104: OS Task Allocation Matrix (Core 2 - Diagnostics & Memory)

Core 2 Task Mapping (QM Diagnostics & NvM):
- OsTask_100ms_Core2 (Priority 5): Dem_MainFunction, Dcm_MainFunction, NvM_MainFunction, ThermalManagementSWC.

Page 105: Non-Volatile Memory (NvM) Block Table

NvM Block Configurations:
- NvM_Block_1 (64 Bytes): Learned Throttle Sensor Calibration (CRC16).
- NvM_Block_2 (512 Bytes): DEM Fault Memory DTC Records (Redundant Block).
- NvM_Block_3 (128 Bytes): Vehicle Coding & Variant Configuration (Write Once).

Page 106: Flash EEPROM Emulation (Fee) Layer

Fee Emulation Strategy:
Emulates physical EEPROM using two 64 KB sectors of on-chip Flash memory. Performs automatic double-buffering and background wear-leveling during low CPU load periods.

Page 107: Software Build and Toolchain Configuration

Build Toolchain:
- Compiler: Tasking VX-toolset for TriCore v6.3r1.
- Optimization: -O3 with MISRA C:2012 checking enabled.
- Linker: Explicit section allocation script placing ASIL code in isolated physical Flash sectors.

Page 108: Integration Verification (MIL, SIL, HIL Testing)

Integration Testing Framework:
- Model-in-the-Loop (MIL): MATLAB/Simulink algorithm validation.
- Software-in-the-Loop (SIL): Virtual RTE execution in Windows/Linux test harness.
- Hardware-in-the-Loop (HIL): dSPACE SCALEXIO real-time simulator verifying 10ms execution deadlines.

Page 109: Technical Glossary & Abbreviations

Glossary of Terms:
- AUTOSAR: Automotive Open System Architecture
- ASIL: Automotive Safety Integrity Level (ISO 26262)
- BSW: Basic Software
- CAN FD: Controller Area Network with Flexible Data-Rate
- CDD: Complex Device Driver
- DCM: Diagnostic Communication Manager
- DEM: Diagnostic Event Manager
- DTC: Diagnostic Trouble Code
- E2E: End-to-End Signal Protection
- HLD: High-Level Design Document
- NvM: Non-Volatile Memory Manager
- RTE: Runtime Environment
- SWC: Software Component
- UDS: Unified Diagnostic Services (ISO 14229)

Page 110: Final Architectural Sign-Off & Approvals

Formal Architecture Sign-Off:
This 110-page High-Level Design specification has been audited and approved as the baseline architecture for Electric Powertrain Domain Controller ECU_EMS_01 Release 1.0.

Approved By:
- Chief System Architect: Dr. E. Schmidt
- Functional Safety Officer: H. Weber
- Chief Engineering Director: Dr. M. Vance